

Suraj Rao

surajrao@umd.edu | +1 2406678312 | College Park, MD | [LinkedIn](#) | [GitHub](#) | [Website](#) | Open To Relocate

EDUCATION

University of Maryland College Park, United States

Sep 2025 - May 2027

Master of Science in Applied Machine Learning [GPA: 3.9/4]

Courses: Machine Learning, Deep Learning, NLP, CV, Generative AI, Statistics, ML Computing Systems, ML Optimization

Indian Institute of Information Technology Pune, India

Aug 2020 - Aug 2024

Bachelor of Technology in Computer Science and Engineering [GPA: 8.09/10]

Courses: Data Structures & Algorithms, OOP, Databases, Image Processing, Cloud Computing, Operating Systems

EXPERIENCE

AI Research Associate Intern | *Mantra Softech, Bangalore, India*

Dec 2023 – Oct 2024

- Built and optimized a video surveillance pipeline by deploying real-time YOLOv8 for people detection and counting
- Reduced model size 4× via quantization, enabling parallel deployment across 4 camera streams using multiprocessing
- Increased throughput from 10 to 35 FPS via frame filtering, batching, and model profiling; deployed on RSC-101 [\[Product\]](#)
- Evaluated DALL-E Mini, Stable Diffusion, CLIP to build a text-to-video retrieval system for ~300+ hours of video footage

AI Research Associate | *IIT Delhi, Delhi, India*

Oct 2024 – Jul 2025

- Contributed to occluded vehicle detection by evaluating transformer-based vision models in Indian road scenarios
- Curated high-occlusion (>50%) data from RSUD20K, IDD, BDD100K via pseudo-labeling and built a Tkinter GUI for false positive correction, reducing annotation time from 7 min to 15 sec (~96%) [\[GitHub\]](#)
- Fine-tuned 4 SOTA transformer detectors (DETR, DINO, Grounding DINO, Co-DETR) on 4 NVIDIA GPUs using distributed multi-GPU training and benchmarked them using mAP and recall

AI Research Assistant | *UMD MIRAGE Lab, Maryland, United States*

Jan 2026 – Present

- Built end-to-end lattice defect detection pipeline using YOLO26 on ~21K images across 3 materials for edge deployment
- Addressed 1:9 class imbalance using data augmentation and PyTorch weighted sampling, improving minority class detection
- Improved thickness classification by analyzing failure cases and refining training strategies, reducing false positives by 45%
- Achieved 96.3% mAP across 6 classes and deployed on NVIDIA Jetson using ONNX and TensorRT for real-time inspection

PROJECTS

Multimodal Semantic Search Engine | *PyTorch, CLIP, FAISS, CUDA, Streamlit, HuggingFace*

[\[GitHub\]](#)

- Built a multimodal semantic search engine using CLIP for text-to-image and image-to-image retrieval on 5.4K+ images
- Generated CLIP embeddings and indexed them with FAISS, improving search speed by 40% and enabling <200 ms latency
- Developed a Streamlit app with text and image query support, returning top-5 semantically relevant results
- Enhanced usability using HTML/CSS and deployed a scalable, containerized pipeline on AWS SageMaker

Smart Crop Yield Prediction | *XGBoost, RandomForest, SVM, Flask*

[\[GitHub\]](#)

- Built a crop yield prediction system for 60+ crops using multi-year agricultural data (2007–2020) across India
- Trained and benchmarked Random Forest, Support Vector Regression, XGBoost using R^2 and MAE with cross-validation
- Achieved peak R^2 of 0.965 and reduced prediction error by 25% (MAE: 0.895 → 0.667) by selecting XGBoost
- Deployed the model via a Flask web application for real-time yield prediction with ~700 ms latency

Multi-Objective YOLOv8 Optimization for Edge Deployment | *Python, ONNX, TensorRT*

[\[GitHub\]](#)

- Built an ONNX based pipeline to export, quantize (FP32/FP16/INT8), benchmark and rank 144 YOLOv8 configs by latency, model size and mAP
- Implemented Pareto-based optimization under deployment constraints; 51 of 144 configurations satisfied the constraints
- Converted the optimal ONNX configuration to a TensorRT engine and deployed it on NVIDIA GPU and Jetson Orin Nano

SKILLS

Programming: Python, C/C++, SQL, HTML, CSS

Libraries: PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, OpenCV, Tkinter, Flask, HuggingFace

Systems: CUDA, ONNX, Distributed Training, Inference Optimization, Quantization, Model Evaluation

AI & ML: LLMs, RAG, Transformers, Agentic AI, Prompt Engineering, Fine-Tuning, OpenAI, Gemini, Claude

Tools: Git, Linux, Docker, Streamlit, FAISS, FastAPI, AWS SageMaker, MLflow, CI/CD, TensorBoard